

Md. Ajmain Istiak Apon

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🔗 onlyapon • 📄 Codeforces: fynm

EDUCATION

University of Dhaka

Dhaka, Bangladesh
Expected Oct 2026

B.Sc. in Robotics and Mechatronics Engineering

Relevant Coursework: Machine Learning, Artificial Intelligence, Intelligent Systems and Robotics, Digital Image Processing and Robot Vision, Digital Signal Processing, Linear Algebra, Statistics for Engineers.

TECHNICAL SKILLS

Languages

Python, C++, C, SQL, JavaScript, LaTeX

ML & Deep Learning

PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers

Computer Vision

OpenCV, Roboflow

LLMs & Agents

LangChain, LangGraph

Backend & APIs

FastAPI, SQLAlchemy, Pydantic, OAuth2 / JWT

Tools & Robotics

Git, Linux, ROS2, STM32, Jupyter, VS Code

EXPERIENCE

Undergraduate Research Assistant

2024 – 2025

University of Dhaka — Funded Research Project: Stomata Detection with YOLOv11

- Trained a **YOLOv11n** model to detect and localize stomata in microscopy images of plant leaves, achieving **0.992 mAP@0.5** and **0.836 mAP@0.5:0.95** (precision 0.969, recall 0.966).
- Curated a custom dataset of **300+ images** annotated in **Roboflow**; built the training pipeline with augmentation, train/validation split, and held-out evaluation.

RESEARCH & PROJECTS

Senior Thesis — Generative Memorability

Jan 2026 – Present

Steering Latent Diffusion Models via Graph-Embedded Spatial Guidance — under faculty supervision

- Integrated AMNet memorability guidance into a Latent Diffusion Model (Stable Diffusion) pipeline to steer generation toward more memorable outputs while preserving fidelity.
- Achieved **+60% memorability uplift** across a 20-prompt evaluation set, independently validated with **ResMem** as an external scorer.
- Extending the pipeline with graph-embedded spatial guidance via custom U-Net modifications and cross-attention steering, benchmarked against Stable Diffusion baselines.

Label-Efficient Medical Segmentation via SSL Volumetric Prediction

Mar 2026 – Present

Independent Research Project

- Designing a label-efficient 3D segmentation pipeline using ViT-Base and Joint-Embedding Predictive Architecture (JEPA), targeting competitive accuracy with only 3–5 annotated volumes.
- Self-supervised pretraining on 1,200+ unannotated CT scans (TotalSegmentator) to learn global spatial geometry via 90% slice masking.
- Built a hardware-constrained training workflow on a single 24GB RTX 3090 using Automatic Mixed Precision (AMP) and gradient accumulation.
- Few-shot fine-tuning that freezes the ViT backbone and trains a 1×1 Conv decoder, benchmarked against fully supervised nnU-Net baselines. *Stack*: PyTorch, MONAI, Nibabel, CUDA.

REST API with JWT Auth (FastAPI)

Apr 2026

- Built a multi-router FastAPI service for an articles/authors blog backend with **SQLAlchemy** ORM and **Pydantic v2** request/response schemas.
- Implemented **OAuth2 password-flow** login issuing **JWT bearer tokens** (bcrypt password hashing via passlib); structured the app with dependency-injected DB sessions and per-resource routers.

COMPETITIVE PROGRAMMING

Codeforces (fynm): 500+ problems solved across dynamic programming, graph theory, greedy, and advanced data structures.

LEADERSHIP

- President**, Robotics & Mechatronics Student Club, University of Dhaka
 - Program Coordinator**, IEEE Robotics & Automation Society, University of Dhaka
- Oct 2025 – Present
Mar 2024 – Apr 2025